the meetings of the National Security Council. I'd carefully configured it to ingest materials at a slow, constant pace, so as not to monopolize the undersea fiber-optic cable tying Hawaii to Fort Meade, but it still pulled so many more documents than any human ever could that it immediately became the NSAnet's most comprehensive readboard.

Early on in its operation I got an email that almost stopped Heartbeat forever. A faraway administrator—apparently the only one in the entire IC who actually bothered to look at his access logs—wanted to know why a system in Hawaii was copying, one by one, every record in his database. He had immediately blocked me as a precaution, which effectively locked me out, and was demanding an explanation. I told him what I was doing and showed him how to use the internal website that would let him read Heartbeat for himself. His response reminded me of an unusual characteristic of the technologists' side of the security state: once I gave him access, his wariness instantly turned into curiosity. He might have doubted a person, but he'd never doubt a machine. He could now see that Heartbeat was just doing what it'd been meant to do, and was doing it perfectly. He was fascinated. He unblocked me from his repository of records, and even offered to help me by circulating information about Heartbeat to his colleagues.

Nearly all of the documents that I later disclosed to journalists came to me through Heartbeat. It showed me not just the aims but the abilities of the IC's mass surveillance system. This is something I want to emphasize: in mid-2012, I was just trying to get a handle on how mass surveillance actually worked. Almost every journalist who later reported on the disclosures was primarily concerned with the targets of surveillance—the efforts to spy on American citizens, for instance, or on the leaders of America's allies. That is to say, they were more interested in the topics of the surveillance reports than in the system that produced them. I respect that interest, of course, having shared it myself, but my own primary curiosity was still technical in nature. It's all well and good to read a document or to click through the slides of a PowerPoint presentation to find out what a program is *intended* to do, but the better you can understand a program's mechanics, the better you can understand its potential for abuse.

This meant that I wasn't much interested in the briefing materials—like, for example, what has become perhaps the best-known file I disclosed, a slide deck from a 2011 PowerPoint presentation that delineated the NSA's new surveillance posture as a matter of six protocols: "Sniff It All, Know It All, Collect It All, Process It All, Exploit It All, Partner It All." This was just PR speak, marketing